Folium: Small-Scale Native App for Community Citizen Science and Botanical Cataloging

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INDUSTRIAL PROJECT

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1 Problem & Objectives

Context & Problem

- Bounded locations — parks, reserves, campuses, urban corridors — host **significant botanical biodiversity** with scarce structured digital documentation.
- **iNaturalist** and **Flora Catalana** offer no offer no location-specific scoping, role hierarchies, or community-tailored gamification.
- No **mobile-first**, **community-driven**, **georeferenced** platform existed for local botanical monitoring.

Objectives

1. Native **mobile app** for submitting georeferenced citizen-science observations.
2. Weighted-consensus **validation system** for botanical data quality assurance.
3. **GDPR** compliance and **production deployment** (Google Play Store).
4. UAB Campus as a pilot "**Living Lab**" for hyper-local biodiversity monitoring.

Benchmarks

iNaturalist — Consensus validation model (Research Grade ≥ 2/3 agreement)

ICO (Catalan Ornithology Institute) — Expert alert system for anomalous observations

Creative Commons BY-NC 4.0 — Licensing framework for all user-generated content

Calendar

January 26	Pivot & Definition
February 26	Core Infra-strure
March 26	Features
April 26	Gamification
May 26	GDPR & Deploy
June 26	Defense

2 Architecture & Methodology

Technology Stack

React Native · Expo SDK 51 · TypeScript · NativeWind · Supabase · PostgreSQL 15 · PostGIS · RLS · EAS Build · Fastlane CI/CD · API · Google OAuth 2.0

Architecture Diagram

CLIENT

1 Presentation Layer — Expo · React Native · TypeScript 5.x

Expo SDK 51+ · TypeScript (strict) · Expo Router (file-based) · NativeWind (Tailwind RN) · expo-location (GPS) · expo-camera (sensor) · react-native-maps · Supercluster (marker clustering) · react-native-reanimated

Navigation tabs: Interactive Map · Species Gallery · Capture & Submit · Social Feed · User Profile

Targets: **Android** (Google Play Store) · **iOS** (Expo Go Tunnel for demo) · Local SQLite cache for offline field sessions

HTTPS / TLS · JWT Bearer Token · Auto-generated REST + WebSocket

2 API Layer — Supabase PostgREST + Edge Functions

PostgREST (auto-generated REST API) · Edge Functions (Deno runtime) · PostgreSQL Realtime (WebSocket) · Row-Level Security enforcement

Edge functions handle: consensus score recalculation · expert alert triggers · account deletion cascade · push notification dispatch

SQL + Row Level Security (RLS) · JWT → user role → policy check

3 Data Layer — PostgreSQL 15 + PostGIS — RLS ON ALL TABLES

PostgreSQL 15 · PostGIS (geography type) · JSONB columns (traits, images) · Row Level Security · EU region (GDPR)

•profiles •species •observations •validations •media_votes •reports •plant_lists

Supabase unified platform · S3-compatible Storage · JWT Auth

4 Storage Layer — Supabase Storage

observation-images
[scientific-name]_[organ]_[uuid].jpg
Public read · Write-protected
Linked to observations.images JSONB

user-profiles
avatar_[user_id].jpg
Fallback: 300×300 pastel emoji canvas
Linked to profiles.avatar_url

5 Identity Layer — Supabase Auth + JWT

Google OAuth 2.0 · Email + Password · JWT (HS256)

JWT payload → `user.role` → RLS policy → table access

Roles: **student** · **expert** · **professor** · **admin**

GDPR: self-serve account deletion via Edge Function cascade

Session: HTTP-only secure cookie / Async Storage

EXTERNAL SERVICES & BUILD PIPELINE

PlantNet API — AI species identification from photo. Returns confidence-ranked GBIF candidates. Advisory only — subject to community validation.

Google OAuth — OAuth 2.0 provider. Tokens exchanged via Supabase Auth. Enables single-tap sign-in without password management.

EAS Build — Expo Application Services. Compiles TypeScript/JS → Android .aab (Gradle) and iOS .ipa (Xcode). OTA updates bypass store review.

Fastlane — CI/CD automation. Uploads binary, screenshots & metadata to Google Play Console. Eliminates manual submission errors.

Google Play Store — Primary distribution channel. Submitted May 2026. Reviewer test account provided. iOS via Expo Go Tunnel (no $99/yr fee).

Observation Submission Data Flow (Field Loop)

1. **Capture** — expo-camera opens · high-res photo taken · expo-location grabs GPS coordinates
2. **AI Suggest** — Photo sent to PlantNet API · confidence-ranked species list returned · user confirms or overrides
3. **Enrich** — User fills morphological traits (JSONB) · organ colors via eyedropper · offline → SQLite queue
4. **Upload** — Image → Supabase Storage (observation-images) · metadata → PostgreSQL (observations table)
5. **Validate** — Community swipes → validations table · confidence_score updated · status: pending → community_verified → scientifically_validated

Total Project Cost: USD $25 · Supabase: free tier · EAS Build: free tier · GitHub: free · Resend (email): free tier · PlantNet: free (academic) · Google Play registration: $25 (one-time) · Apple Developer Program: excluded ($99/yr)

Validation Model

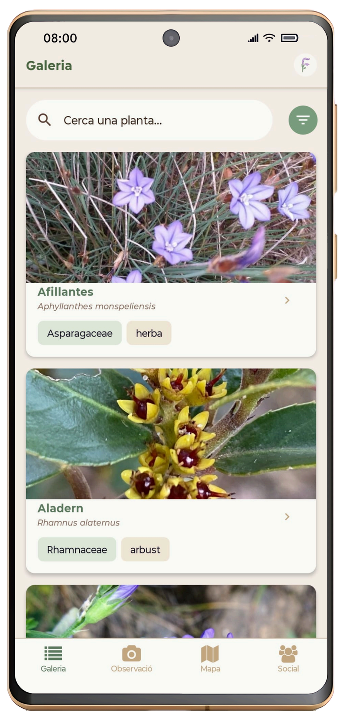
pending → comunity_verified → scientifically_verified · archived

Student	+5 pts	−5 / report
Expert	+25 pts	−15 / report
Professor	+50 pts	instant validation ✓

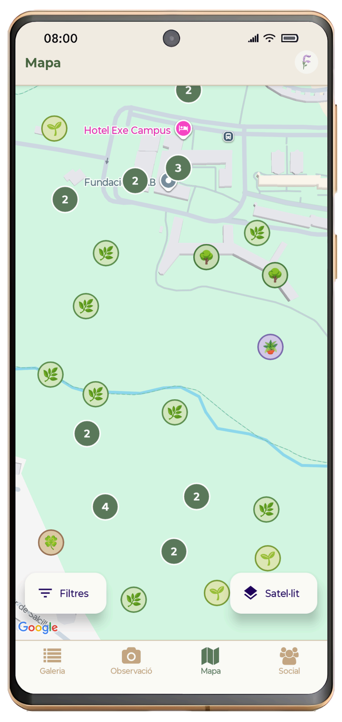
`community_verified` *if score ≥ 50 · auto-archived if ≤ −60*

3 Results & Conclusions

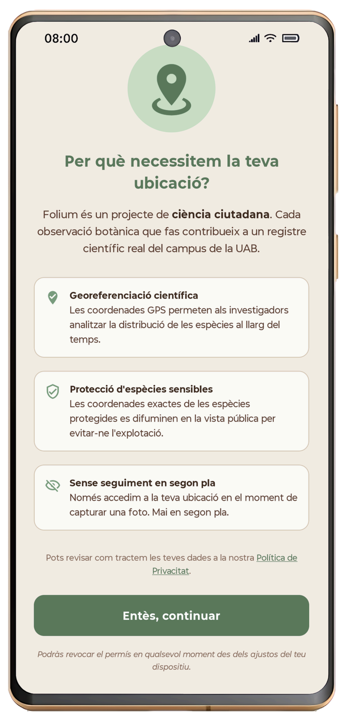
App Screenshots



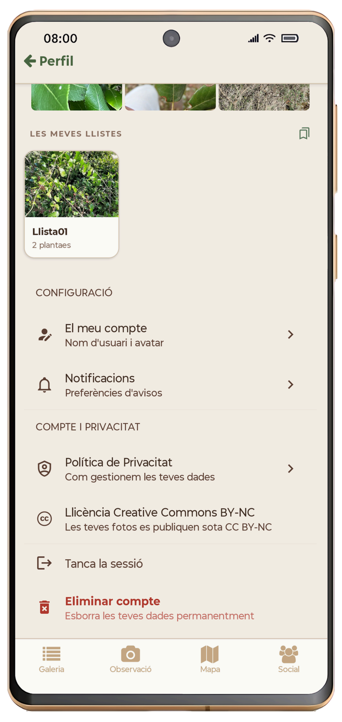
Gallery



Map



GPS disclosure



Account & Privacy

App Demo

Key Achievements

- App published on **Google Play Store** (May 2026)
- **Creative Commons BY-NC 4.0** licence for all UGC
- Consensus validation engine (iNaturalist-inspired, rule ≥ 50 pts)
- Push alerts to experts for anomalous observations (Edge Functions)
- Full GPS permission onboarding for fieldwork
- Total operating cost: **$25 USD** (Google Play, one-time)

Conclusions

Folium demonstrates that a single developer in a university setting can deliver a **production-quality citizen-science platform**. The Living Lab model at UAB Campus is viable as a scalable approach for hyper-local biodiversity monitoring.

Open challenge: pilot data collection pending Play Store approval (June 2026).

Key Words

Citizen Science · Botanical Cataloging · Supabase · GDPR · Mobile Deployment · Biodiversity Monitoring · React Native

References

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2. Fastlane Documentation. *docs.fastlane.tools*
3. Creative Commons. *creativecommons.org*
4. Expo / React Native docs. *docs.expo.dev*
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Future Work

- **Expanded role hierarchy** — Finer-grained ranks for deeper community engagement and contribution incentives
- **Project organizations** — Delimit geographic scope and species dataset per community or initiative
- **Seasonal Wrapped** — Annual activity summaries per user to boost long-term retention